Feature Survival and Intervention Preservation under Multilingual Weight Quantization

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Abstract

Does a feature used to explain a language model still have the same effect after the model's weights are rounded to fewer bits? We test this in 27 languages, mainly using Gemma-2-2B and a fixed sparse autoencoder (SAE). We first reproduce an earlier study and identify a missing beginning-of-sequence token that substantially changes its Gemma results. With the corrected input protocol, removing selected language features harms next-token prediction more than removing matched controls in all 27 languages. This pattern also appears in Gemma-2-9B. Effects remain close to full precision at six bits and grow at four bits under simple rounding; GPTQ reduces this change. Activation changes alone give an incomplete picture: under legacy RTN at four bits, only 13.2% of a 1,487-feature sample passes the activation-survival cutoff, yet 99.7% can still be identified from their output effects. Activation correlation improves prediction of effect similarity by a small but consistent 3.1–4.4%. We also find no clear group effect on a reading-comprehension test and little repair from restoring the selected features. Together, the results show which aspects of feature behavior survive quantization and why activation measurements and intervention tests should be reported together.

1 Introduction

Quantization rounds a language model's weights to a smaller set of values. It can reduce storage and computation costs, but it can also change the model's behavior. Those changes can differ across languages and tasks (Marchisio et al., 2024). We ask what quantization does to the internal features used to explain a multilingual model.

A sparse autoencoder (SAE) is a separate network that describes a model's hidden state using a small number of active features from a larger learned dictionary. A feature has an activation strength and a direction in the hidden state. Some features activate more often for one language than for others. Earlier work has identified and edited such language features (Deng et al., 2025). Other work measures whether their activation patterns survive weight rounding (Duan, 2026).

These measurements leave a practical question open. If a feature's activation changes, does its effect on the model also change? A feature may still respond most strongly to French, for example, while an edit to that feature has a different effect on output predictions. Conversely, its activation pattern may change while its output effect remains recognizable. We measure both activations and the effects of edits.

The main study uses Gemma-2-2B, a fixed Gemma Scope SAE (Lieberum et al., 2024), and 27 languages. We compare two round-to-nearest (RTN) implementations with GPTQ (Frantar et al., 2022). We remove, weaken, strengthen and restore selected features, then measure changes in text prediction and reading comprehension. We also test how the same edit behaves when the hidden state or later model weights change. Fresh Wikipedia articles provide a second text source. Gemma-2-9B and a smaller Llama study extend parts of the investigation.

Four questions organize the completed work:

RQ1: Which language features retain their activation patterns after quantization, and how does the quantizer affect this?

RQ2: How does editing language features affect text prediction and comprehension before and after quantization?

RQ3: How well do activation measurements predict whether a feature's output effect is preserved?

RQ4: How much does the same feature edit change when the model is quantized, and what explains that change?

We chose these final questions to organize the accumulated evidence. The original hypotheses, choices fixed before each analysis, and recorded outcomes remain separate.

The study makes three main contributions. First, it establishes controlled effects of language-feature groups on next-token prediction across two Gemma sizes. Second, it shows that changed activations can coexist with recognizable output effects, and measures the modest predictive link between them. Third, it tests how much of an edit's change comes from the earlier hidden state, later weights, or sensitivity shared with other edit directions. The BOS audit also identifies and corrects an important problem in the input protocol inherited from earlier work.

2 Related Work

Quantization and language features. Multilingual quantization effects depend on language and evaluation task (Marchisio et al., 2024). Deng et al. (2025) identify language-associated features and show that editing them changes full-precision models. Duan (2026) measure activation agreement before and after RTN quantization. Our work connects these questions by comparing activation changes with the measured effects of feature edits across languages and quantizers.

What SAE features can tell us. SAEs are useful tools, but their features need not form a unique or complete list of the model's computations. Features can split a concept across several units or combine several concepts in one unit (Leask et al., 2025). Sparsity choices also affect what the SAE separates (Chanin and Garriga-Alonso, 2026). We therefore keep the dictionary fixed and measure the effect of each specified edit. This gives a clear paired comparison without requiring every feature to represent a complete, isolated mechanism.

Testing effects directly. Makelov et al. (2025) evaluate whether features support control on a task without requiring exact recovery of a true feature inventory. SAEBench shows that improvements in proxy measures need not improve practical performance (Karvonen et al., 2025). Work on activation patching also studies where quantization errors affect model outputs (Chang et al., 2025). These studies motivate our use of several outcomes and control edits. Each outcome answers a specific question: activation agreement, preferred language, output-effect similarity or task performance.

3 Experimental Setup

3.1 Models, text and language features

Our main model is Gemma-2-2B (Gemma Team et al., 2024). We use a Gemma Scope JumpReLU SAE (Rajamanoharan et al., 2024) with 16,384 features at the residual stream after layer 12. The residual stream is the hidden state passed between model layers. Model inference uses bfloat16; SAE calculations and edits use float32.

The FLORES-200 panel (NLLB Team et al., 2022) contains the same sentence content in 27 languages, with nine languages in each descriptive resource tier (Appendix A). We select features on the development split. Activation measurements use 27,036 content tokens per language from devtest; prediction-loss experiments use 300 aligned devtest sentences per language. The tiers group languages by resources; they do not isolate the effect of model training exposure.

A feature is language-selective when it fires much more often for one language than its average across languages. If $p_{\ell f}$ is the firing rate of feature f in language ℓ, we measure this as

$$S_f = \frac{\max_\ell p_{\ell f}}{\mathrm{mean}_\ell\, p_{\ell f}}. \tag{1}$$

Selection uses a shuffled-language cutoff (95th percentile, 50 permutations), at least 50 peak-language tokens, a panel firing rate of at least 10^{-5}, and the same peak language in at least 80 of 100 resamples. This selects 8,086 features. Of these, 6,663 have a unique supported peak language on FP devtest. FP denotes the unquantized, full-precision reference model.

For each language we edit its 64 selective features with the highest mean development-set activation. We call this group K64. We compare it with nonselective control groups matched on firing rate and mean activation, both on a log scale. The K64 loss and graded-scaling experiments use three control draws; the saved manifest contains up to five. Matching is imperfect, particularly for Thai, Yoruba and Lao. The resulting estimate compares these specific edits and controls.

3.2 The BOS problem and our correction

Our first reproduction disagreed strongly with Duan's Gemma results. We then ran their released code and reproduced their reported survival rates

closely: 81.19% versus 81.26% at seven bits, and 51.15% versus 51.30% at six bits. The disagreement led us to identify a methodological problem in that Gemma evaluation: the input windows omitted the beginning-of-sequence (BOS) token.

On the same 256-content-token WikiText windows, adding BOS and excluding its position from feature statistics reduced FP perplexity from 458.4 to 16.1. INT6 survival increased from 51.15% to 97.52%; INT7 survival rose to 99.97%. Lower perplexity means better next-token prediction. Because each regime defines its own active-feature population, these survival rates do not track exactly the same feature set.

We corrected the input protocol for subsequent experiments: prepend one BOS token, then exclude BOS and padding from statistics and edits. The audit establishes a large input-protocol effect on the reported Gemma measurements. It does not determine the validity of every result in Duan's paper. We also retired our first pipeline after finding separate implementation flaws; its findings are excluded. The audit and correction history remain in the evidence record.

3.3 Quantizers and additional tests

RTN rounds each weight to the nearest allowed value. Our original implementation, called legacy RTN, shares a scale across corresponding coordinates of attention heads. Matched RTN uses one scale per output channel and the same rounding grid as GPTQ. This separates changes in the rounding setup from the additional benefit of GPTQ.

GPTQ uses calibration examples to compensate for rounding error (Frantar et al., 2022). We use 8,192 Wikipedia tokens per language, or 221,184 in total. Calibration text is separate from the evaluated sentences and articles. GPTQ changes both calibration and error compensation, so their individual contributions are not isolated.

All quantized weights remain stored and evaluated as floating-point values. We test the effect of rounding weights, rather than integer-kernel speed or memory savings. Activation and loss comparisons cover eight to four bits (INT8–INT4); additional intervention tests focus on INT6 and INT4. INT16 and INT12 provide small weight-change references. INT16 changes some weights and is not an identical rerun of FP arithmetic.

The reading-comprehension test uses Belebele (Bandarkar et al., 2024): 385 held-out questions per language in 18 languages that passed calibration. Its main outcome measures the probability assigned to the correct option, converted to a loss. We report answer accuracy separately. A second transfer panel uses fresh Wikipedia articles in six languages. Although its articles differ from GPTQ calibration articles, both sources are Wikipedia.

The Gemma-2-9B extension uses its layer-20, 16k Gemma Scope SAE and the same 27-language, 300-sentence scaling design under legacy RTN. We checked the imported raw-file hashes and regenerated the analysis byte-identically; we did not repeat GPU inference. The Llama-3.2-1B check uses a layer-12 TopK SAE with 14,336 features and $k = 48$. Its model, dictionary, tokenizer and training differ together, so it tests another setup without isolating the cause of any difference.

3.4 How we report uncertainty

Most analyses are exploratory. They reuse FLORES sentences or follow questions suggested by earlier results. Fixing a later analysis before seeing its outputs improves transparency, but does not make reused data independent confirmation. The original held-out Belebele test family is confirmatory. The present reporting revision adds no model run or statistical test.

Bracketed ranges are 95% bootstrap intervals unless stated otherwise. Bootstrap resampling estimates uncertainty by repeating the analysis on resampled data. Group-loss tests resample paired sentences while keeping languages and feature groups fixed. The reviewed 9B–2B comparisons resample paired languages. Feature-effect prediction tests hold out one language at a time and resample languages for intervals. Transfer tests hold out feature identities and resample features and source sentences. These intervals describe different sources of uncertainty; they do not measure variation over all possible models or SAE dictionaries.

4 What We Measure

4.1 Activations, language preference and output effects

We keep three questions separate. First, does a feature activate similarly on the same tokens? Its activation correlation is

$$R^b_{1f} = \mathrm{corr}_t(a^{\mathrm{FP}}_f(t), a^b_f(t)). \qquad (2)$$

The earlier study's survival rule calls a feature preserved when this correlation exceeds 0.9. Second, does the feature still fire most strongly for the same

language? We test this only where FP establishes a unique supported peak language. Third, does editing the feature still change the output in the same way?

To edit a feature, we change its activation and add the corresponding decoder direction to the hidden state. For residual state x, activations a and decoder directions d_f,

$$x' = x + \sum_{f \in G} (a'_f - a_f) d_f. \tag{3}$$

Setting $a'_f = 0$ removes the feature; setting $a'_f = \alpha a_f$ weakens or strengthens it. A donor patch replaces its activation with the FP value at the same token. This edit keeps the SAE's reconstruction error in the hidden state. Identity and sham edits check that the edit path itself does not change outputs.

Prediction loss, or negative log-likelihood (NLL), measures how much probability the model assigns to the actual next token. Lower loss is better; we report it in nats/token. Let $h^b_\ell(G)$ be the loss increase after editing group G in language ℓ and precision state b. We subtract the average increase from matched controls:

$$E^b_\ell = h^b_\ell(G_\ell) - \frac{1}{R} \sum_{j=1}^{R} h^b_\ell(C_{\ell j}). \tag{4}$$

We call E excess harm. A positive value means removing the language group hurts more than removing its controls. The effect ratio $E^b_\ell / E^{\mathrm{FP}}_\ell$ compares this harm with FP: one means the same effect, and 1.5 means 50% greater harm from the specified edit. We distinguish the median and mean of language ratios from the ratio of mean effects; small FP denominators can make them differ.

For individual features, we record how removal changes the model's output logits, the scores assigned to possible next tokens. This vector is the feature's effect fingerprint. Cosine similarity with its FP fingerprint, denoted W_1, measures how closely the output-change directions agree; one is identical direction. The F5 experiment samples 1,487 features. It compares each feature with itself, other features and matched control directions. A stricter similarity cutoff of 0.9489 comes from the INT16 reference. Passing this cutoff means close agreement with that reference; it is a different test from identifying which feature produced an effect.

4.2 Separating activation shrinkage from other changes

A correlation does not show every activation change. If every activation shrinks by the same factor, correlation can remain one. We therefore write centered quantized activations b as a scaled FP signal plus a residual:

$$b = ga + e, \quad g = \frac{\mathrm{Cov}(a, b)}{\mathrm{Var}(a)}, \quad \mathrm{Cov}(a, e) = 0. \tag{5}$$

Here g is gain: values below one indicate shrinkage. The residual e is the part that scaling alone does not explain; calling it noise does not imply it is random hardware error. For cells with nonzero variances,

$$r^2 = \frac{g^2 \sigma_a^2}{g^2 \sigma_a^2 + \sigma_e^2}, \tag{6}$$

$$\lambda = \log \frac{\sigma_e^2}{g^2 \sigma_a^2} = \log \sigma_e^2 - 2 \log |g| - 2 \log \sigma_a. \tag{7}$$

Thus agreement depends on residual change relative to the FP signal's size. The sign of r follows g.

We also measure changes before the SAE's activation threshold is applied. For pre-activation z_f and learned threshold θ_f, define

$$\nu_f = \frac{\sqrt{\mathrm{mean}_{t: a^{\mathrm{FP}}_f(t) > 0} (z^b_f(t) - z^{\mathrm{FP}}_f(t))^2}}{\theta_f}. \tag{8}$$

This expresses change in units of that feature's threshold, using tokens where it was active in FP. We separately count activations that switch off. Running totals recover these measures without storing all token-by-feature arrays; silent features and cases where the formulas are undefined remain counted in the population records.

4.3 Testing the same edit in changed contexts and weights

An edit's output effect can change for two reasons. The earlier layers may deliver a different hidden state, or the later layers may respond differently to the edit. We cross FP and quantized states before and after the intervention layer to separate these contributions. The edit direction and strength remain fixed at their FP values.

Let $D[p, s]$ be the mean centered output-logit change, where p selects the earlier model state and

s the later weights. Each cell subtracts its own unedited output. With Q denoting quantized,

$$C = D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{FP}], \tag{9}$$

$$W = D[\mathrm{FP}, \mathrm{Q}] - D[\mathrm{FP}, \mathrm{FP}], \tag{10}$$

$$I = D[\mathrm{Q}, \mathrm{Q}] - D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{Q}] + D[\mathrm{FP}, \mathrm{FP}]. \tag{11}$$

C measures the earlier-state contribution, W the later-weight contribution, and I their interaction. Their vectors add exactly to the total change. Vector lengths and medians do not generally add. Group experiments instead calculate the corresponding contrasts directly in excess prediction loss. A fifth condition uses the quantized model's own activation strength, allowing us to compare a fixed edit with its native feature removal.

Control edits use a matched different feature direction and three random orthogonal directions with the same norm. These test how much change is shared with other directions. Each direction has an INT16 reference. We retain real-direction items whose FP output effect exceeds five times that reference change: 2,203 of 6,464 feature–context pairs, covering 235 of 404 features. Relative transfer error is

$$T = \frac{\|D[\mathrm{Q}, \mathrm{Q}] - D[\mathrm{FP}, \mathrm{FP}]\|}{\|D[\mathrm{FP}, \mathrm{FP}]\| + \mathrm{floor}}. \tag{12}$$

The floor is the INT16 reference scale. Lower T means the fixed edit transfers more closely. Prediction tests split feature identities into calibration and held-out halves. Source sentences can occur in both halves, so holding out features does not also hold out documents.

5 Results

5.1 RQ1: Preservation depends on feature activity and quantizer

The initial tests did not establish that language-selective features consistently survive worse than controls. Under legacy RTN, selective-feature survival was lower by 0.28 and 1.44 percentage points at INT7 and INT6. None of the three original tests passed Holm correction for multiple comparisons (adjusted $p = 0.514, 0.128$ for these two contrasts). This leaves small differences possible. Selecting only more strongly language-selective features ($S > 5$) gives a larger INT6 gap of -7.16 points. Matching activation magnitude in ten bins reverses it to $+4.58$. The result depends strongly on which features are compared.

Low-, mid- and high-resource languages have legacy INT6 survival rates of 69.4%, 90.6% and 95.1%. The tier association becomes smaller after accounting for activation size. Later measures also show tier patterns at the small INT16 reference perturbation. GPTQ's adjusted tier result changes with the adjustment choice. The measured differences are real descriptions of these feature populations, but do not by themselves show that low-resource language mechanisms suffer greater damage.

GPTQ clearly improves preservation in this setting. At INT4 it raises the mean per-language survival rate by 17.83 points $[16.29, 19.38]$ over legacy RTN and reduces the NLL increase by 0.461 nats/token $[0.348, 0.595]$. Both improve in all 27 languages. Compared with matched RTN, the gains are 11.86 survival points $[11.18, 12.50]$ and 0.255 nats/token less loss increase $[0.193, 0.327]$. These intervals resample paired languages. Matched RTN itself improves on legacy RTN. Its small selective-feature contrast crosses the significance threshold, while GPTQ's does not. We therefore find that feature activity and quantizer choice shape preservation; a consistent penalty for language-selective features is not established across methods.

5.2 RQ2: Language features matter for text prediction

Removing a language's K64 group harms its text prediction more than removing matched controls in all 27 languages. In FP Gemma-2-2B, mean excess harm is 0.704 nats/token $[0.684, 0.723]$ and the median is 0.262. Every language's sentence-bootstrap interval is above zero. Weakening the features gradually gives the same conclusion: median harm rises from 0.022 to 0.090, 0.169 and 0.262 as the activation multiplier falls from 0.354 to 0.177, 0.088 and zero. This ordering holds in all 27 languages.

The removal effect remains near FP at six bits and grows at four bits. Legacy RTN's median effect ratio is 1.03 $[1.02, 1.04]$ at INT6 and 1.54 $[1.43, 1.62]$ at INT4. At INT4 the mean ratio is 2.14, while the ratio of mean effects is 1.33. GPTQ reduces the median ratio to 1.22 $[1.14, 1.24]$. A larger removal effect means the quantized model is more sensitive to that edit; it does not mean its language ability improved.

Gemma-2-9B shows the same broad RTN pattern (Table 1). FP excess harm is positive in all 27 languages, and the INT4 ratio exceeds one in 26. The paired INT4 median difference between model sizes is $+0.048$ $[-0.777, 0.838]$, so we cannot re-

9B original RTN	Estimate	95% interval
FP median harm	0.270	[0.197, 0.608]
INT6 median ratio	1.019	[0.978, 1.045]
INT4 median ratio	1.586	[1.238, 2.264]
INT4 mean ratio	2.479	[1.781, 3.368]
INT4 ratio of means	1.474	[1.263, 1.937]

Table 1: Reviewed Gemma-2-9B group effects. FP harm is in nats/token; ratios compare excess ablation harm with FP. All intervals here resample languages. Different ratio aggregations answer different questions. No 9B GPTQ result is included.

solve a size effect. The result extends the group pattern to the larger Gemma model. Two defects in the original prediction checklist are documented in Appendix B; we retain the measured curves without counting those defective items as independent confirmations.

Llama shows a smaller FP median effect, 0.048 $[0.018, 0.173]$, with positive effects in 21 of 27 languages. It does not meet the predefined consistency rule, which required 24 languages, and its INT4 pattern differs from Gemma. Since both the model and SAE change, the comparison demonstrates variation between setups without identifying its cause.

The comprehension test gives a different answer. On held-out Belebele, removing the language group does not produce a clear extra correct-option loss: -0.0001 nats/item $[-0.0035, +0.0031]$ in FP. Baseline accuracy is about 42.6%; accuracy is a separate measure from this loss. The planned retention tests require a positive FP effect, so they cannot be interpreted when that requirement fails. Larger random nonselective edits reduce accuracy by 1.2–2.2 points; edits to other languages' selective features reach 6.0 points. Sham edits leave outputs unchanged. The test detects larger interventions, while small effects of K64 remain possible. GPTQ also shows no clear K64 contrast at INT4, where accuracy is above chance. The next-token finding therefore does not automatically extend to reading comprehension.

Restoring these features repairs little of the loss. Replacing K64 activations with FP values recovers 2.6% $[2.4, 2.8]$ of the legacy INT4 loss gap. Restoring the full layer residual recovers 39.3% $[38.5, 40.0]$. GPTQ gives 2.0% $[1.6, 2.3]$ and 41.4% $[40.2, 42.4]$, respectively. These percentages measure what the particular patches repair. They cannot be added as shares of the cause of quantization damage.

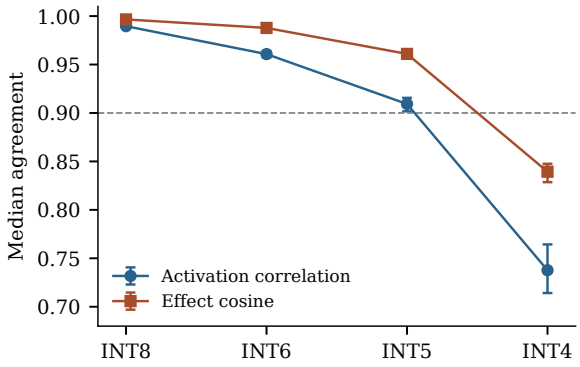


Figure 1: Activation correlation and effect-fingerprint cosine on the same 1,487-feature legacy-RTN sample. Points are medians; bars are 95% language-bootstrap intervals. The dashed line is the historical activation-survival cutoff, not a common decision rule for both metrics. Different trajectories motivate measuring both outcomes.

The original feature-strengthening study does not meet its repair rules. A follow-up on 562 fresh sentences finds a small legacy-RTN benefit from multiplying K64 activations by 1.1: -0.00135 nats/token $[-0.0020, -0.0003]$ (97.5% interval; 20 of 27 languages improve). The two primary multipliers use Holm correction; the INT4-versus-FP comparison also meets its rule. This is a small improvement. GPTQ does not repeat the fine-gain trend on the main 300-sentence sample; the fresh 562-sentence experiment was legacy-only.

5.3 RQ3: Changed activations can retain recognizable effects

Activation agreement, preferred language and output effects do not change together. Among features with an established FP peak language, 70.8% retain it at legacy INT4. That question differs from whether activations remain highly correlated on the same tokens.

The same-sample comparison is especially clear. Of 1,487 F5 features, only 13.2% pass activation correlation 0.9 at legacy INT4. Yet their median output-effect cosine is 0.839 $[0.829, 0.848]$: 99.9% exceed the different-feature control, and 99.7% match their own FP identity most closely. Only 0.3% meet the much stricter INT16-based similarity cutoff. The output effects have changed, but remain distinctive (Figure 1). GPTQ raises median INT4 effect cosine to 0.923, compared with 0.875 for matched RTN and 0.839 for legacy RTN. Other control directions also drift, so the change is not all unique to SAE features.

Activation correlation still provides useful information. For 1,412 features with reliable reference effects, we predict effect similarity while holding out one language at a time. The baseline uses feature activity, FP effect size and context counts. Adding activation correlation reduces prediction error by 3.1–4.4% across all six quantizer/precision conditions; every interval is above zero (Table 2). This is a small, consistent gain, below the project's 10% practical target. The Llama INT4 gain is similar, 4.2% $[1.6, 6.7]$; INT6 remains uncertain.

Keeping the same peak-language label is less helpful in the tested classifications. Its descriptive discrimination score is 0.506 for legacy INT4 and about 0.49 for GPTQ using the legacy reference labels, both near chance. This binary label answers a different question from continuous prediction with activation correlation.

The activation audit helps explain why a single correlation is incomplete. It covers 214,506 FP-active language–feature pairs, reproduces earlier correlations within 3×10^{-8}, and checks Equation 6 to about 10^{-14} where defined. Across six conditions, the rank correlation between λ and FP activation spread is -0.507 to -0.534. For threshold-scaled perturbation ν, it is only -0.004 to 0.114. Threshold scaling reduces dependence on activation spread, though dependence on peak activation remains (correlations 0.25–0.41).

At INT4, median gain is 0.594, 0.672 and 0.823 for legacy RTN, matched RTN and GPTQ. The corresponding shares of previously active tokens switched off are 50.0%, 42.9% and 27.0%. These measures reveal shrinkage and switching that correlation alone does not describe. Adding gain and ν after correlation changes prediction improvement by -0.4 to $+1.8\%$, with every interval including zero. They clarify the changes without establishing additional predictive benefit here.

5.4 RQ4: The same edit transfers more closely under GPTQ

The fixed edit and the quantized model's own feature removal give different effects. For K64 at legacy INT4, the fixed FP-strength edit has median excess harm 0.848 nats/token, compared with 0.263 in FP. Using quantized activation strengths reduces harm to 0.433. In the crossed experiment, median absolute contributions are 0.310 from earlier context, 0.104 from later weights and 0.071 from their interaction. At INT6, context and weight terms are both about 0.009.

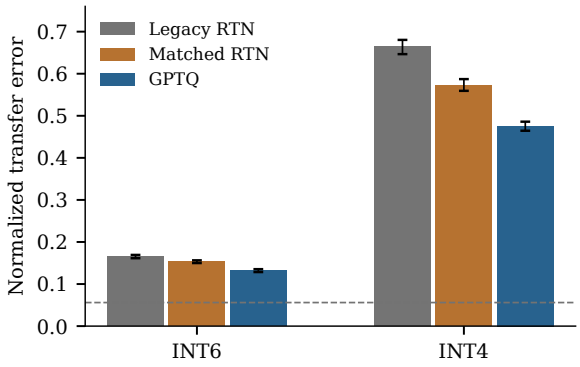


Figure 2: Median normalized transfer error by quantizer and precision on identical eligible items. Bars show 95% crossed-bootstrap intervals for each arm. GPTQ preserves the tested intervention more closely, with remaining error above the INT16 reference.

For individual output vectors, the later-weight term is larger than the context term: relative norms are 0.560 and 0.359, with interaction 0.315 at legacy INT4. This differs from the group-loss result because output-vector length and prediction loss measure different aspects of an edit. The term sizes describe the specified crossed experiment, rather than a unique partition of all model damage.

GPTQ preserves the same fixed interventions more closely (Table 3, Figure 2). At INT4, median cosine rises from 0.768 under legacy RTN to 0.822 under matched RTN and 0.877 under GPTQ. Relative error falls from 0.664 to 0.572 and 0.476. The comparisons use the same eligible feature–context pairs. Figure intervals describe each method; overlapping intervals alone do not test a paired difference.

Sensitivity shared with other directions explains a substantial part of the transfer change. Using control-direction sensitivity improves prediction of legacy INT4 transfer error by 21.5% $[12.7, 29.4]$ beyond the strongest activation/gradient baseline. At INT6, the estimated gain is 10.9% $[-1.2, 21.4]$, so the original rule requiring utility at both widths is not met. A similar gain at INT16 indicates sensitivity to weight perturbations more generally.

Fresh Wikipedia articles repeat the INT4 quantizer ordering: median cosine is 0.754, 0.808 and 0.876 for legacy RTN, matched RTN and GPTQ. Control-direction sensitivity improves prediction at both widths for the RTN methods. For GPTQ it meets the rule at INT4, but not INT6 (3% $[-14, 17]$), where remaining error is about 1.4 reference floors. GPTQ calibration uses different Wikipedia articles, so shared domain may

Quantizer	Precision	Add activation correlation	Add gain and ν after correlation	Features
Legacy RTN	INT6	4.4 [2.6, 6.2]	-0.1 [-1.6, 1.4]	1,412
Legacy RTN	INT4	3.7 [2.3, 5.2]	1.8 [-0.3, 3.7]	1,412
Matched RTN	INT6	3.8 [1.6, 5.8]	-0.4 [-1.6, 0.9]	1,412
Matched RTN	INT4	3.1 [1.8, 4.4]	1.0 [-0.5, 2.5]	1,412
GPTQ	INT6	4.1 [1.9, 6.2]	-0.2 [-1.9, 1.2]	1,412
GPTQ	INT4	3.5 [2.1, 5.0]	0.7 [-1.5, 2.8]	1,412

Table 2: Reduction in mean absolute prediction error (MAE), in percent, with 95% language-bootstrap intervals. Adding activation correlation gives a small positive gain in every condition. Adding gain and threshold-scaled change ν after correlation gives no clear further improvement. All rows use 1,412 features.

State	Quantizer	Cosine	Rel. error
INT6	Legacy RTN	0.984	0.165
	Matched RTN	0.986	0.153
	GPTQ	0.990	0.132
INT4	Legacy RTN	0.768	0.664
	Matched RTN	0.822	0.572
	GPTQ	0.877	0.476

Table 3: Median fixed-intervention transfer on the same eligible real-direction items. The legacy INT16 small-weight-perturbation reference has cosine 0.998 and relative error 0.056. Table values summarize changed effects; they do not imply loss of feature identity.

contribute to its advantage.

Averaging across contexts also hides variation. At legacy INT4, comparing averaged fingerprints raises cosine by 0.126 $[0.121, 0.131]$ relative to averaging context-level cosines. The difference is 0.0145 $[0.011, 0.017]$ at INT16. Reports should therefore state whether an effect is preserved on average or in individual contexts.

6 Discussion and Conclusion

The selected language groups contribute to next-token prediction in all 27 languages of both Gemma sizes. Their removal effects remain near FP at six bits and increase at four bits under RTN. GPTQ reduces that increase and better preserves activation patterns and output effects in the main 2B study. These are controlled findings about the edits and models tested.

The results also explain why calling a feature simply "surviving" or "damaged" loses information. Many features fail the activation cutoff while retaining distinctive output effects. Activation correlation still supplies a small positive predictive gain. Measuring shrinkage, switching and the same edit under changed contexts gives a fuller account of preservation.

Some proposed effects are not established. The initial selective-vulnerability tests do not give a consistent result across quantizers, the comprehension test shows no clear K64 effect, and the tested repairs recover little loss. These outcomes define the useful reach of the methods alongside the positive text-prediction findings.

Mechanistic interpretability faces shared problems of mixed features, incomplete dictionaries and task-dependent explanations. We address them by fixing the dictionary, defining edits precisely and comparing several outcomes and controls. Broader model and dictionary studies would test generalization. The present evidence already supports specific, reproducible findings about multilingual feature edits under quantization.

Limitations

Models and quantization. Most evidence comes from Gemma-2-2B at one layer with one SAE. Gemma-2-9B extends the RTN group results; Llama tests a narrower, different setup. The findings may change with other models, layers or dictionaries. We round weights in floating-point inference, so the study does not measure integer-kernel deployment or activation quantization. Gemma-2-9B GPTQ is pending and contributes no result. Further 9B quantizer and arithmetic-precision studies would extend the present evidence.

Features and edits. SAE features can mix or split concepts, and large edits can produce unusual hidden states. Our estimates apply to the specified feature directions and control groups. Activity matching and alternate-direction controls do not equalize every feature property. Fixed FP-strength edits also differ from native feature removal. Restoring one part of an activation does not measure its additive share of all quantization damage.

Data and uncertainty. Most analyses reuse examined FLORES data. The F5 sample is stratified,

and prediction splits hold out languages or features rather than all shared documents. Bootstrap intervals describe the stated resampling units while keeping other choices fixed. They do not cover uncertainty across model training seeds or all later analysis choices. The six-language external panel is small, and GPTQ's calibration and external evaluation share a domain.

Reproduction and access. The 9B review reproduces analysis from supplied raw files, rather than repeating GPU inference. Appendix B records two defects in its original prediction definitions. Code, frozen analysis choices, tables and provenance are retained in the repository. Large raw outputs are in a private artifact dataset and currently require author sharing for independent access. The correction history, including our retired pipeline and BOS audit, is retained.

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A Language Panel

Each tier has nine languages. High: English, German, Spanish, French, Portuguese, Russian, Chinese, Japanese and Korean. Mid: Hindi, Bengali, Tamil, Turkish, Vietnamese, Indonesian, Thai, Polish and Ukrainian. Low: Kannada, Telugu, Yoruba, Sundanese, Amharic, Igbo, Hausa, Khmer and Lao. These groups summarize language resources; they do not isolate training exposure.

The sentences are parallel, but their token counts differ across languages. Equal-token activation tests and aligned-sentence loss tests therefore use different token populations. The 1,487-feature F5 sample, its 1,412-feature reliable subset and the eligible transfer pairs are also different populations. Their rates should not be pooled as repeated measurements of one quantity.

B 9B Review and Prediction Definitions

The 9B study uses the same graded-scaling analysis as 2B. Review checked file hashes, precision labels and sentence identifiers, then regenerated curves and reports byte-identically. Three informative predictions are supported: positive FP excess harm, near-FP effects at INT8/INT6, and larger effect ratios at INT4.

Two original criteria need separate treatment. First, the inequality for graded suppression was written backwards. Its literal order holds in zero languages; the intended order, increasing harm with stronger suppression, holds in all 27. We report both the definition error and the observed curves. Second, a criterion asks for selectivity above 1.3 after features were already selected above a higher cutoff. It is guaranteed by selection and supplies no independent confirmation. These defects affect the prediction checklist, not the reproduced effect estimates. Paired model-size comparisons do not resolve a difference between 2B and 9B.

C Reading Thresholds and Intervals

The project uses several decision conventions: activation correlation above 0.9, near-reference effect cosine above 0.9489, transfer effects above five reference floors, and a 10% prediction-improvement target. These make decisions repeatable, but estimates remain useful when they miss a target. For example, the 3.1–4.4% correlation-prediction gains have intervals above zero even though they fall below 10%. Adding gain and ν gives intervals including zero, a different outcome. A feature can also miss the close-similarity cutoff while remaining identifiable from its fingerprint.

The activation audit counts 214,506 FP-active language–feature pairs. Depending on condition, 4–239 pairs have zero or otherwise undefined variances and are excluded from the gain/noise identity check. Silent and clipped cases remain counted in population tables. Reproducing correlations and checking the identity test numerical consistency; they do not establish that one measure captures all of a feature's function.

D Reproduction and Evidence Map

This manuscript adds no model run or hypothesis test. Figures use committed CSV and JSON estimates with their saved intervals. The evidence map records each result's source, population, uncertainty and analysis status. Original hypotheses, fixed analysis choices, generated verdicts and dated histories remain available separately.

The main 2B quantizer comparisons and activation audit are complete. Gemma-2-9B GPTQ remains pending and appears in no result table. Further models, dictionaries and deployment tests would extend the study. They are not required to report the controlled comparisons already completed.